

EXPERIMENT NO: 13

Determination and plotting of intrinsic impedance and phase velocity of various medium

Aim:

To determine and plot intrinsic impedance and phase velocity of various medium.

Software required:

- SCILAB version 6.0.1

Program:

```
EXP 13.sci (C:\Users\user\OneDrive\Documents\EXP 13.sci) - SciNotes
File Edit Format Options Window Execute ?
EXP 13.sci (C:\Users\user\OneDrive\Documents\EXP 13.sci) - SciNotes
EXP 13.sci X
7
8 //Constants
9 uo = 4 * %pi * 10^(-7);
10 eo = 8.854 * 10^(-12);
11
12 //Intrinsic impedance
13 n = sqrt(uo * ur / (eo * er));
14
15 //Phase velocity
16 pv = 1 / sqrt(uo * ur * eo * er);
17
18 //Display results
19 mprintf("\nIntrinsic Impedance = %.6f Ohm\n", n);
20 mprintf("Phase Velocity = %.6e m/s\n", pv);
21
22 //Values for plotting
23 x = linspace(0, 10, 50);
24
25 y = n * ones(1, length(x));
26 s = pv * ones(1, length(x));
27
28 //Create figure
29 clf(1);
30 clf();
31
32 //Intrinsic impedance graph
33 subplot(2, 1, 1);
34 plot2d3(x, y);
35 xlabel("Distance");
36 ylabel("Intrinsic Impedance (Ohm)");
37 title("Intrinsic Impedance");
38
39 //Phase velocity graph
40 subplot(2, 1, 2);
41 plot2d3(x, s);
42 xlabel("Distance");
43 ylabel("Phase Velocity (m/s)");
44 title("Phase Velocity");
45
46 xgrid();
47
```

Output:

Scilab 2026.1.0 Console

File Edit Control Applications ?

File Browser

C:\Users\User\Documents\

Name

- Documents
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- My Pictures
- My Videos
- ex 6.sci
- EXP 13.sci~

File/directory filter

Scilab 2026.1.0 Console

Enter relative permeability μ_r : 3×10^{-6}

Enter relative permittivity ϵ_r : 6×10^{-6}

Intrinsic Impedance = 251.156206 Ohm

Phase Velocity = 8.327657e+06 m/s

-->

Variable Browser

Name	Value	Type	Visibility	Memory
ans	1x1	Graphic ...	local	216 B
eo	8.85e-12	Double	local	216 B
er	54	Double	local	216 B
n	251	Double	local	216 B
pv	8.33e+06	Double	local	216 B
uo	1.26e-06	Double	local	216 B
ur	24	Double	local	216 B
x	1x50	Double	local	608 B
y	1x50	Double	local	608 B

Command History

News feed

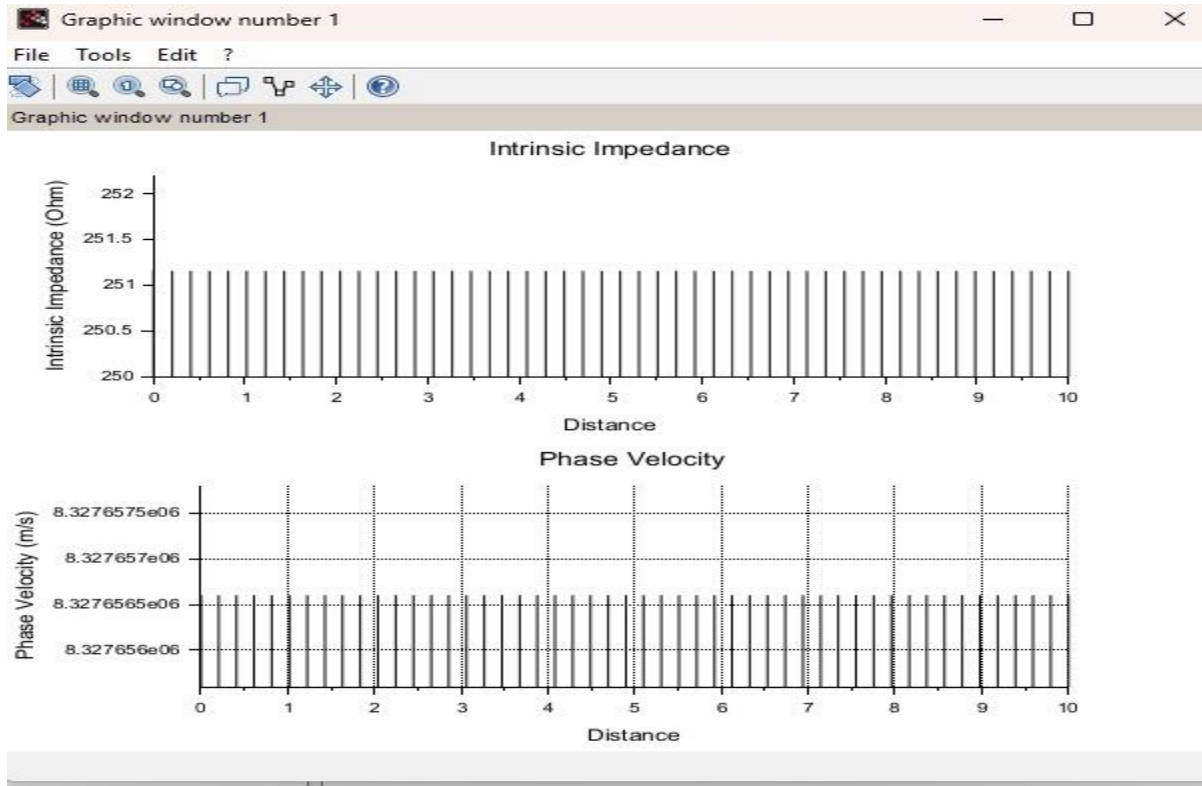
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5:26 AM 8/24/2026



Theoretical calculation:

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Determination and plotting of intrinsic impedance and phase velocity of various medium.

Output :

Enter μ_r value = 3×10^{-6}

Enter ϵ_r value = 6×10^{-6}

Practical Output :

" Intrinsic Impedance = 251.156206 Ohm "

" Phase Velocity = $8.327657 \times 10^6 \text{ m/s}$ "

Theoretical Calculation :

Intrinsic Impedance : $\eta = \sqrt{\frac{\mu}{\epsilon}}$

Taking approximately,

$$\mu = 3 \times 10^{-6} \text{ H/m}$$

$$\epsilon = 4.76 \times 10^{-11} \text{ F/m}$$

Therefore,

$$\eta = \sqrt{\frac{3 \times 10^{-6}}{4.76 \times 10^{-11}}}$$

$$\eta = \sqrt{63025.2}$$

$$\eta = 251.05 \Omega$$

Phase velocity :

$$v = \frac{1}{\sqrt{\mu \epsilon}}$$

$$v = \frac{1}{\sqrt{(3 \times 10^{-6}) (4.76 \times 10^{-11})}}$$

$$v = 8.37 \times 10^6 \text{ m/s}$$

$$v = 8.33 \times 10^6 \text{ m/s}$$

Intrinsic Impedance = 251.05Ω

phase velocity = $8.33 \times 10^6 \text{ m/s}$

Case 1: To determine the intrinsic impedance and phase velocity of electromagnetic waves in free space(vacuum).

Program:

```
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File Edit Format Options Window Execute ?
EXP 13.sci (C:\Users\user\OneDrive\Documents\EXP 13.sci) - SciNotes
EXP 13.sci X
1 clc;
2 clear;
3
4 // Constants
5 mu0 = 4 * pi * 10^(-7); // Permeability of free space (H/m)
6 epsilon0 = 8.854 * 10^(-12); // Permittivity of free space (F/m)
7
8 // Calculate intrinsic impedance
9 eta0 = sqrt(mu0 / epsilon0);
10
11 // Calculate phase velocity
12 vp = 1 / sqrt(mu0 * epsilon0);
13
14 // Display results
15 mprintf("Intrinsic Impedance (eta0) = %.6f Ohms\n", eta0);
16 mprintf("Phase Velocity (vp) = %.6e m/s\n", vp);
17
```

Output :

SciLab 2026.1.0 Console

File Edit Control Applications ?

File Browser

C:\Users\user\Documents\

Documents

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- EXP 13.sci

File/Directory filter

SciLab 2026.1.0 Console

Intrinsic Impedance (eta0) = 376.734309 Ohms
Phase Velocity (vp) = 2.997956e+08 m/s

--> |

Variable Browser

Name	Value	Type	Visibility	Memory
epsilon0	8.85e-12	Double	local	216 B
eta0	377	Double	local	216 B
mu0	1.26e-06	Double	local	216 B
vp	3e+08	Double	local	216 B

Command History

```
// -- 10/08/2026 14:33:17 -- //
// -- 10/08/2026 14:50:34 -- //
// -- 24/08/2026 04:27:35 -- //
// -- 24/08/2026 04:29:27 -- //
3*10^-7
6
4
2
5
3
4
5
3
4
```

News feed

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Case 2: To determine the intrinsic impedance and phase velocity for a wave propagating through a conductive medium like copper.

Program:

```
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File Edit Format Options Window Execute ?
EXP 13.sci (C:\Users\user\OneDrive\Documents\EXP 13.sci) - SciNotes
EXP 13.sci X
1 clc;
2 clear;
3
4 // Constants
5 f = 3e9; // Frequency (Hz)
6 mu = 4 * pi * 10^(-7); // Permeability (H/m)
7 sigma = 5.9e7; // Conductivity of copper (S/m)
8 epsilon_0 = 8.854e-12; // Permittivity of free space (F/m)
9 epsilon_r = 1; // Relative permittivity
10
11 // Imaginary unit
12 j = 1i;
13
14 // Effective permittivity
15 epsilon = epsilon_0 * epsilon_r;
16
17 // Angular frequency
18 omega = 2 * pi * f;
19
20 // Intrinsic impedance
21 eta = sqrt((j * omega * mu) / (sigma + j * omega * epsilon));
22
23 // Phase velocity
24 vp = 1 / sqrt(mu * epsilon);
25
26 // Display results
27 disp("Intrinsic Impedance eta (Ohms):");
28 disp(eta);
29
30 disp("Phase Velocity vp (m/s):");
31 disp(vp);
32
```

Output :

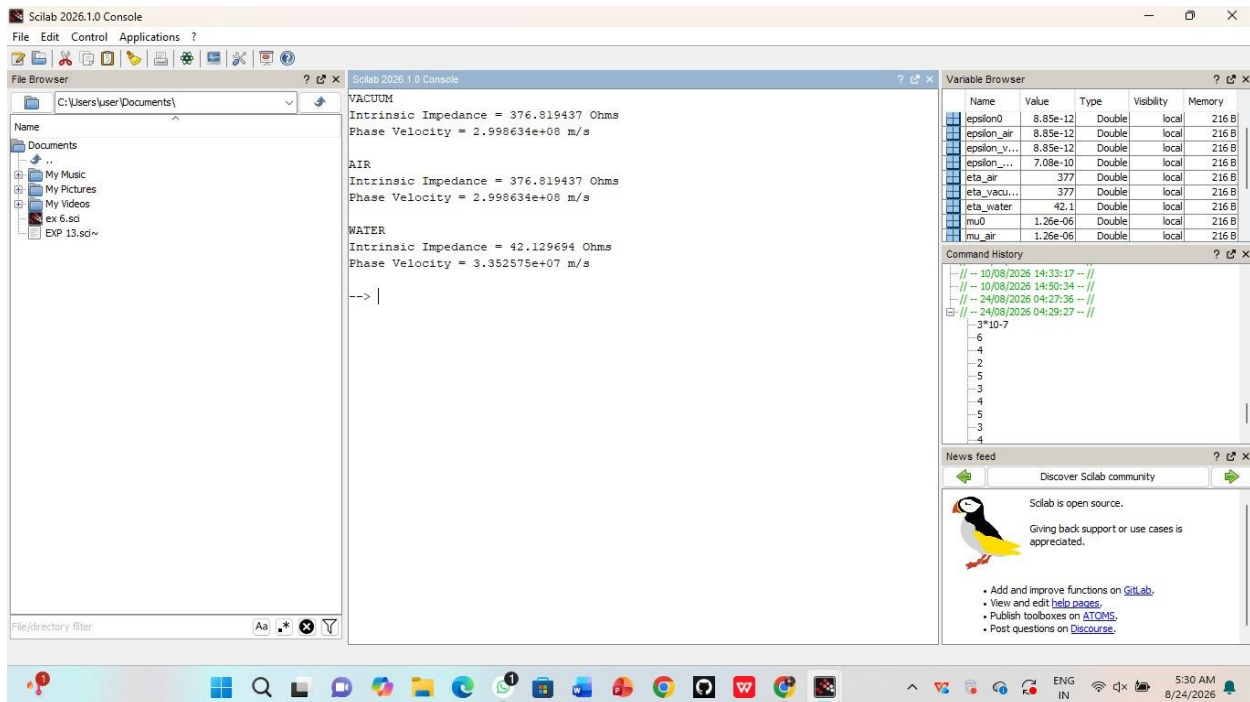
```
Scilab 2026.1.0 Console
File Edit Control Applications ?
C:\Users\user\Documents\
Name
Documents
My Music
My Pictures
My Videos
ex 6.sci
EXP 13.sci
File/directory filter
Aa + - X
Scilab 2026.1.0 Console
" Intrinsic Impedance eta ( Ohms) : "
0.01428999 + 0.01428991i
" Phase Velocity vp (m/s): "
2.998D+08
-->
Variable Browser
Name Value Type Visibility Memory
epsilon 8.85e-12 Double local 216 B
epsilon_0 8.85e-12 Double local 216 B
epsilon_r 1 Double local 224 B
eta 0.0143 + 0.0143i Double local 216 B
f 3e+09 Double local 224 B
j 0 + 1i Double local 216 B
mu 1.25e-06 Double local 216 B
omega 1.88e+10 Double local 216 B
sigma 5.8e+07 Double local 216 B
Command History
-- // -- 10/08/2026 14:33:17 -- //
-- // -- 10/08/2026 14:50:34 -- //
-- // -- 24/08/2026 04:27:36 -- //
-- // -- 24/08/2026 04:29:27 -- //
3*10^-7
6
-4
-2
-3
-4
-5
-3
-4
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```

Case 3: To determine and compare the intrinsic impedance and phase velocity of electromagnetic waves propagating through different media: Vacuum, Air, and Water.

Program:

```
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File Edit Format Options Window Execute ?
EXP 13.sci (C:\Users\user\OneDrive\Documents\EXP 13.sci) - SciNotes
EXP 13.sci X
1 clc;
2 clear;
3
4 // Constants
5 mu0 = 4 * pi * 10^(-7); // Permeability of free space (H/m)
6 epsilon0 = 8.85 * 10^(-12); // Permittivity of free space (F/m)
7
8 // =====
9 // VACUUM
10 // =====
11 mu_vacuum = mu0;
12 epsilon_vacuum = epsilon0;
13
14 // =====
15 // AIR
16 // =====
17 mu_air = mu0;
18 epsilon_air = epsilon0;
19
20 // =====
21 // WATER
22 // =====
23 mu_water = mu0;
24 epsilon_water = 80 * epsilon0;
25
26 // Function to calculate properties
27 function [eta, vp] = calculate_properties(mu, epsilon)
28     eta = sqrt(mu / epsilon); // Intrinsic Impedance (Ohm)
29     vp = 1 / sqrt(mu * epsilon); // Phase velocity (m/s)
30 endfunction
31
32 // =====
33 // VACUUM
34 // =====
35 [eta_vacuum, vp_vacuum] = calculate_properties(mu_vacuum, epsilon_vacuum);
36
37 fprintf("VACUUM\n");
38 fprintf("Intrinsic Impedance = %.6f Ohms\n", eta_vacuum);
39 fprintf("Phase Velocity = %.6e m/s\n", vp_vacuum);
40
41 // =====
42 // AIR
43 // =====
44 [eta_air, vp_air] = calculate_properties(mu_air, epsilon_air);
45
46 // =====
47 // WATER
48 // =====
49 [eta_water, vp_water] = calculate_properties(mu_water, epsilon_water);
50
51 fprintf("WATER\n");
52 fprintf("Intrinsic Impedance = %.6f Ohms\n", eta_water);
53 fprintf("Phase Velocity = %.6e m/s\n", vp_water);
54
```

Output:



Result:

Thus the determination and plotting of intrinsic and phase velocity is performed and successfully verified in copper.